TABLE OF CONTENTS

Chapter 1 Introduction
Introduction....3
Annual Water Report Principles....4
Mission Statement and Objectives....5
Provincial Requirements....6

Chapter 2 Existing Infrastructure
Supply....7
Distribution....8

Chapter 3 2010 Evaluation
Source Water....9
Distribution....11
Water Treatment....12
Customer Service and Accountability....12
Water Demand Management....13

Chapter 4 Accomplishments
Source Water....15
Distribution....15
Water Treatment....15
Staff Training....16
Water Demand Management....17

Chapter 5 Improvement Strategy
Improvement Strategy 2011....18
Goals for the Future....19

Glossary....20

Appendix A
2010 Water Sample Range Report....21

Appendix B
2010 Analysis of Water Samples....31

Appendix C
2010 Town of Ladysmith Annual Report for Small Drinking Water Systems....40

INTRODUCTION

The 2010 Annual Water Report continues to demonstrate the Town of Ladysmith's effort towards achieving excellence through continued responsible construction, operation, monitoring, evaluation and management of its water system.

The Town currently meets all of its requirements under water regulations and endeavours to continually improve the water system and public awareness to meet changing needs of the community. The report also identifies areas for improvement which will increase the dependability and sustainability of our water system. Improvements in water initiatives including advanced water treatment, increased community awareness, enhanced customer service programs and more aggressive demand management projects are all major areas of improvement presented within this report.

This report is a working document for water system operations and management assisting the Town in working towards its established goals and mission statement. By increasing public awareness of the water system and service, this report will enable the community to participate in demand management strategies in an informed way. With the understanding and support of the community, the Town can work towards meeting set objectives of enhanced quality, quantity and efficiency.

Water safety is of utmost importance to the Town of Ladysmith. The supply of good, clean drinking water has been taken for granted by the general public in Canada until the events such as the Walkerton E. Coli outbreak brought the safety of the water supply into the public eye. These concerns were brought home to the citizens of the Town of Ladysmith in July, 2001, when the Arbutus Reservoir, the main treated reservoir serving the Town, was subject to a contamination event of unknown cause.

Although these events have brought the issues into the forefront of the public interest, the Town of Ladysmith has long recognized the need for significant upgrades to their existing water supply infrastructure, and have been proactively planning major upgrades to the water supply infrastructure for several years. To ensure an adequate supply of water for now and into the future, the Town commissioned a study to assess the geotechnical and hydrological conditions of the water supply lakes. The study also presented a comparison of the options considered to increase the reliability and quantity of the available surface water supply. A further study was undertaken to determine the feasibility of the Town tying into the Cassidy Aquifer.

Chapter One

ANNUAL WATER REPORT PRINCIPLES

1. This report is an annual review and recommendation document encompassing all operations and management of the water system to ensure accountability to the community for services provided.

2. The mission statement, goals and targets presented in this report will be reviewed annually to provide clear direction for the water systems operation and management for the community.

3. This report will provide direction for improvement strategies for the present and future. The report will be a working document which has flexibility for change from ongoing input, updated research, evaluation and monitoring improvements.

4. The targets presented in this report are monitored each year to measure performance of the water system and if targets cannot be properly monitored to measure success, will be revised to provide effective performance management targets.

5. The improvement strategies, which identify current and future work projects, will be incorporated into their perspective budgets. The information provided to the public will be presented in a comprehensive and understandable form and will be consistent with operation and management budgets within the municipalities five year financial plan.

6. Water reports will contain financial information of the water system to ensure that financial resources are consistent with infrastructure and service planning.

7. This report will be distributed within the community to increase public support for water system improvement initiatives and to encourage public engagement on water conservation.

MISSION STATEMENT & GOALS

Mission Statement

The Town of Ladysmith will work to achieve excellence through continued responsible construction, operation, monitoring, evaluation and management of its water system.

Objectives

- To ensure an adequate supply of high quality water to the community.
- To effectively treat the raw water to provide potable water of integrity to the community.
- To ensure the adequate delivery of high quality potable water delivery to all points within the system for domestic and emergency purposes.
- To ensure effective management of all water system aspects and provide excellent customer service and information to the community.
- To manage water demand by effectively assessing and managing water losses from leakage in the system.
- To develop an effective water conservation program for operations and the wider community.
- To maintain water rates that encourage conservation and resource awareness while providing quality accessible water to consumers.

Chapter One

PROVINCIAL REQUIREMENTS

All drinking water in the water system should meet the Canadian Guidelines for Drinking Water Quality. In British Columbia, the Ministry of Health regulates water suppliers through the Drinking Water Protection Act. This legislation ensures safe drinking water in the province. It requires that the water supplier monitor the drinking water source and distribution system to ensure compliance with the Drinking Water Protection Regulations and report all results to the Health Authority. Water monitoring, inspection and testing, emergency response planning, cross connection control and security standards are all regulated for persons operating a water system.

Changes in water systems must be approved by Vancouver Island Health Authority (VIHA), and conform to the Town's standards and specifications which meet the American Water Works Association (AWWA) standards. In addition, all works must conform to Town bylaws.

Under the BC Water Act, the Town must acquire licenses for withdrawal from water bodies and pay annual dues for the amount of water consumed.

Under the Community Charter the Town may by bylaw regulate, prohibit, and impose requirements in relation to municipal service and public health. The Town must make reports available to the public on request regarding fees imposed under this section.

Chapter Two

EXISTING INFRASTRUCTURE

Supply

The Town of Ladysmith's water is drawn from two distinct sources, which provide water through separate facilities at the north and south ends of the Town. The majority of the Town at the north end is supplied by the Holland Lake and Holland Creek watershed which enters the water supply system at a diversion point at the Chicken Ladder Dam, a small stilling basin located approximately 2.5 km. from the Public Works yard. From this point the water is piped directly to the newly constructed enclosed Arbutus Reservoir, where it is chlorinated and distributed by gravity to North Ladysmith and the Diamond Improvement District.

The southern portion of the Town is supplied by an intake and pipeline from Stocking Lake, which also provides water for Saltair within the Cowichan Valley Regional District. The pipeline carries flow for Ladysmith to the Balancing Reservoir where the high pressure in the pipeline is released and a small amount of storage is provided. From there flow is piped to the chlorination and Contact Tank area and then piped to the distribution system at the south end of the Town of Ladysmith.

The Arbutus Reservoir and Contact Tank top water levels provide the static hydraulic grade lines for the north and south portions of the Town respectively. There is a 2700 m long, 200 mm diameter Asbestos Cement water main connecting these two points which is used, at times of poor water quality in Holland Creek, to fill the Arbutus Reservoir. Thus Stocking Lake becomes the only water source at these times. With increased activity in the Holland Creek watershed since 2000, records indicate that the Stocking Lake source is required for longer frequencies with longer periods of poor quality water in the Holland Creek watershed.

EXISTING INFRASTRUCTURE

Distribution

The topography of the Town of Ladysmith slopes steeply towards the waterfront, resulting in a large range in the water pressures within the Town's water system. With the completion of the new covered Arbutus Reservoir and subsequent increase in distribution pressure of 32 psi in 2008, the Town introduced two specific pressure zones within the system with the installation of four pressure reducing stations.

The distribution system for the Town consists of over 61 km of water main of varying sizes from 100mm to 250 mm. The mains are made of caste iron piping, asbestos cement piping or PVC piping. There are approximately 150 commercial, 3 industrial and 2700 residential water services connections. The caste iron (approximately 1 km.) and the asbestos cement (approximately 20 km.) piping are located mainly in the older section of Town and within the area that was amalgamated from Saltair in 1985. The majority of the asbestos cement piping is 100 mm in diameter and needs to be upgraded and upsized to allow for adequate flows throughout the system. The Town has been actively working on a replacement program since 2000 where a section of asbestos cement piping is replaced with proper sized PVC pipe annually.

Fire hydrants are situated throughout the distribution system at a spacing to offer adequate fire flows for all properties.

Chapter Three

2010 EVALUATION

Source Water

Over the past several years, the Town of Ladysmith has carried out considerable work to evaluate the condition and capacity of the Town's existing water supply. Initial work consisted of an evaluation of the existing storage reservoir dams to determine the feasibility and cost of upgrading of the dams and increasing storage volumes to provide possible additional withdrawals for the Town's water supply to determining the feasibility of accessing the Cassidy Aquifer for additional supply. The evaluation of Prevost Lake, which supplied approximately two weeks of water to the north end of Ladysmith around the beginning of June, identified the need for some major dam repairs to make the dam safe for use. The cost of the repairs was deemed to be excessive for the volume of water provided and the dam was decommissioned in 2009.

A contamination event during the spring of 2006 prevented the Town from using its main source on Holland Creek for an extended period of time. A diversion pipeline from Holland Lake to Stocking Lake was investigated to make available additional storage volume to the limited Stocking Lake storage system. Normally, Stocking Lake is only used by north Ladysmith during periods of the winter and spring, when the turbidity and/or colour of the Holland Creek source is above 1.0 NTU and 15 TCU, respectively, and water demands are low. Work was completed in 2010 which upgraded the chlorine delivery system to allow for adequate chlorine delivery in the south end at high flows such as during a fire event. The Town also installed an online turbidity meter to provide for continuous turbidity monitoring in both the south end and at the Arbutus Reservoir

A review of the annual water use records of the past five years identified that the Town has been using progressively more water from Stocking Lake, from 36% of the total supply in 2002, to 61% of the total in 2007. This is a result of extended periods of high turbidity and colour in the Holland Creek supply. A watershed water quality review of the Holland Creek watershed identified a number of possible causes of the increase in turbidity and colour duration and in 2009 the Town developed a Water Quality Monitoring Plan, which included implementation of a long-term sampling program in the watershed, and initiated other protocols to address potential watershed and associated water quality issues. In 2008, the usage shifted back closer to historical volumes with 41% coming from Stocking Lake and 59% from Holland Creek. 2009 continued the shift with 35% coming from Stocking Lake and 65% from Holland Lake. More severe storms in 2010 brought the usage back up to 55% from Stocking Lake.

The new elevation of the Arbutus Reservoir allows for the entire distribution system to be serviced from one location. This allows the Town to look at centralized treatment which in turn will assist the Town in achieving its goal of reducing its carbon footprint. A grant application was submitted for works that included constructing a dual pipeline from the South End Chlorinator to the Arbutus Reservoir, phase I of the centralized treatment plant, and a new power supply. The application was unsuccessful. Council did authorize the detail design of the pipeline and centralized treatment process to go ahead without the a grant. The design showed that the elimination of the South End Chlorinator provides the ideal location for the Towns first in-line turbine for power generation. A hydraulic Energy Recovery Report was completed and adopted by Council.

The Town also completed a Watershed Protection Plan which identified the location and number of various testing protocol to ensure that the water supply for the Town was maintained at the highest quality. Regular sampling of the water supply started in mid February of 2010.

Chapter Three

2010 EVALUATION

Distribution

As part of an ongoing water main replacement program, the Town replaced the 100 mm asbestos cement water main on High street between 2nd Avenue and Third Avenue with a 150 mm PVC water main. The Town also completed replaced the 200 mm asbestos cement watermain with a 200 mm PVC watermain on Second Avenue between Buller Street and High Street. An additional 28 single family lots, 2 multi family lots and a 40 unit trailer park were developed at the Northwest end of Town in Lot 108.

In July, 2008, Council implemented a toilet rebate program with a $20,000 budget allocation which provided $75 to any resident of Ladysmith that replaced an existing toilet with an approved low flow toilet to a maximum of two toilets per residence. The program was to be completed in December 2009. The program was very well received and Council increased the budget and extended the program to December 31, 2010. A total of 2000 rebate requests were received by the end of December 2010.

A low flow toilet uses approximately 6,600 less gallons of water annually per person than the old standard toilets.

2010 EVALUATION

Water Treatment

The higher elevation of the new Arbutus Reservoir provides an additional 32 psi pressure throughout the distribution system and allows the utility operators to chlorinate the water prior to entering the reservoir which allows for more contact time for the chlorine prior to consumption. This improves water quality and delivers a higher calibre product to customers.

The higher elevation also allows the Town to distribute water from the Arbutus Reservoir to the entire Town. This eliminates the need for a second reservoir at the south end and allows the Town to work towards a centralized treatment facility. The Town submitted an application to the Province for a grant to allow for the construction of a dual pipeline from the South End Chlorinator to the Arbutus Reservoir, the construction of the first phase of a centralized treatment facility, an upgrade to the power supply. The application was unsuccessful. Council authorized the detail design of the work in 2010. The design was completed and included a Hydraulic Energy Recovery component which was adopted by Council.

Customer Service and Accountability

The Town of Ladysmith is committed to delivering high quality water in an accessible, safe, and sustainable manner. The ongoing water quality monitoring and the updated Waterworks Emergency Preparedness Report and the development of the annual water reporting system have all contributed to the high level of customer service and satisfaction in 2010.

Staff have been trained in chlorine handling, Workplace Hazardous Materials Information Systems, and Environmental Operators Certification Program training.

Customers are always welcome to provide feedback on the service they have received, by contacting the Public Works Department.

Development of a water system Standard Operating Procedures manual is underway and preventative maintenance scheduling program is in place to improve water system operations and increase the integrity of the system. Along with these programs there is need for an enhanced data management program and training.

Council approved an Increasing Block Rate water payment structure in 2009. The fourth quarter billing included both the single rate and the increasing block rate. The new billing standard was put into effect in January 2010.

Chapter Three

2010 EVALUATION

Water Demand Management

The Town of Ladysmith has presented water demand management information to the public through events, Council presentations, appliance incentives, and water metering. Residents of Ladysmith have a keen interest in the health and security of their watershed, as was made clear during the Community Visioning sessions held during 2008. Awareness-building activities have proven effective in raising the public consciousness about water consumption.

In the spring of 2008, the Town introduced a low flow toilet rebate program where residents could submit their receipts for up to two toilets and receive a rebate to help offset the costs of their dual flush or low flow toilets. The program was extended till the end of 2010 due to its popularity.

All water customers of the Town of Ladysmith are connected to metered water systems. This helps track usage and provides an incentive for users to reduce their consumption in order to save money. The metering system is also effective in demand management by detecting leaks more quickly than un-metered systems. Waterworks employees and customers are more diligent with tracking their usage over time and will identify abnormally high quantities quickly after a leak has occurred. Water metering has had a positive effect on water demand management. Usage has been reduced from 1,720,171 cubic metres in 2002 to 1,313,950 cubic metres in 2010. This 23.6% decrease in water consumption occurred despite a 20.4 % increase in population.

The Town of Ladysmith is proud to be a leader in operational water conservation. The Increased Block Water Rates introduced by the Council in 2009 will continue to assist in the Towns sustainability goals.

ENCLOSED ARBUTUS RESERVOIR

Chapter Four

SOURCE WATER ACCOMPLISHMENTS

- **Completion of Watershed Protection Plan**
  The Watershed Protection Plan was competed by the Towns Consulting Engineers and adopted by Council. The plan has been submitted to VIHA

- **Completion of the Water Quality Monitoring Program.**
  The Water Quality Monitoring Program was completed and adopted by Council. The plan identifies various locations within the watershed for testing of the raw water and the number and types of tests proposed. Testing started in February, 2010.

- **Turbidity Analysers**
  The installation of turbidity analysers was completed for both the Arbutus Reservoir location and the South End Chlorinator. The installation allows for continuous reads at both locations.

DISTRIBUTION ACCOMPLISHMENTS

- **Replacement of Water Main at High Street and Second Avenue**
  As part of our regular watermain replacement program, crews replaced the asbestos cement water main on High Street between Second and Third Avenue and also on Second Avenue between High Street and Buller Street..

- **Long Term Planning**
  A number of reports were initiated in 2008 to help the Town ensure a safe and adequate water supply for the present and into the distant future. These reports were completed by Koers & Associates in 2009 and subsequently adopted by Council.

WATER TREATMENT ACCOMPLISHMENTS

Centralized Water Treatment

The construction of the enclosed Arbutus Reservoir at a higher elevation allowed the Town to supply treated water from the one location to the entire town. Council authorized the detail design for the dual pipeline between the south end chlorinator and Arbutus Reservoir, Phase I of the centralized treatment and an upgraded power supply to be completed in 2010.

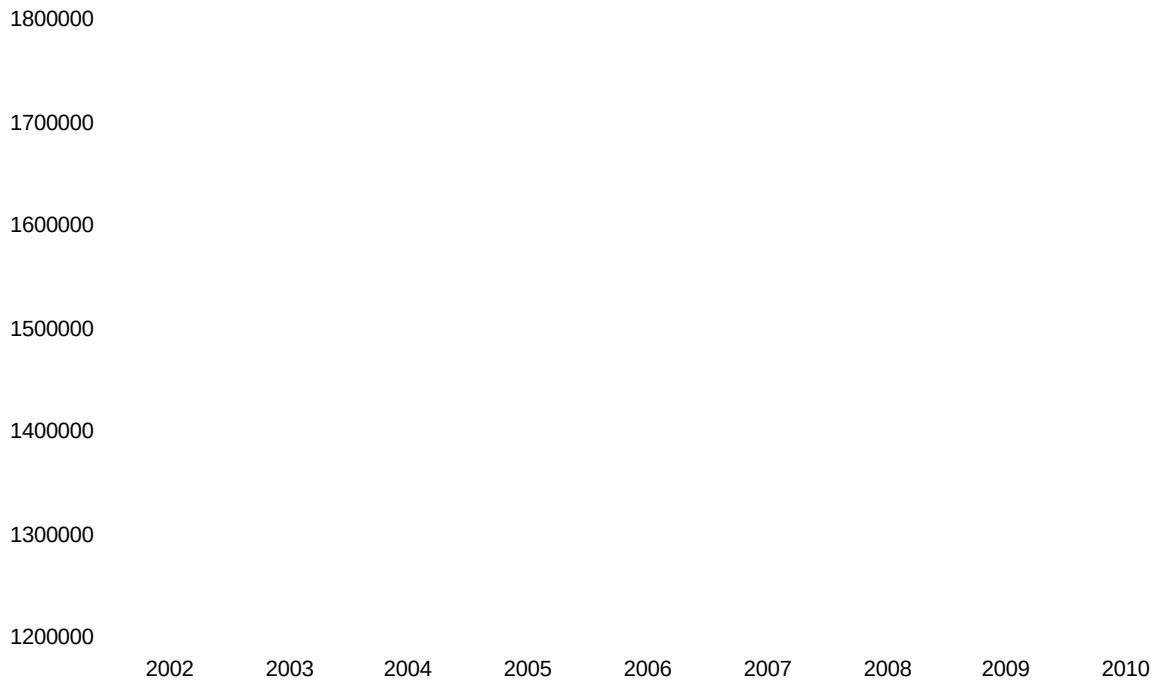
ACCOUNTABILITY AND CUSTOMER SERVICE ACCOMPLISHMENTS

- **Staff Training**
  Training in chlorine handling, Workplace Hazardous Materials Information Systems, and Environmental Operators Certification Program has improved the knowledge and skill of staff. Additionally, there have been training sessions in customer service, leadership, team dynamics, communication, and self and public awareness.

Chapter Four

WATER DEMAND MANAGEMENT ACCOMPLISHMENTS

- **Reduced Water Consumption**
  Although 2009 showed a slight increase in water consumption due to a very dry summer, the previous six consecutive years of reduced consumption and an increased population is an indicator of the success of the Town's water demand management initiatives. The consumption in 2010 was the lowest since annual records were being kept.



- **Low-flow Toilet Rebate**
  The low-flow/dual flush toilet rebate program was extended to the end of 2010. The program has been an effective way to engage residents in water demand management. A total of 311 rebates have been issued since the program was put in place in the spring of 2008.

- **Public Awareness Building**
  Awareness activities at events, publications on water conservation, and public sessions such as the Community Visioning have all contributed to high public awareness of the watershed and the resource of water itself.

IMPROVEMENT STRATEGY 2011

- **Increased training for Waterworks Staff with Environmental Operators Certification Program**
  Continual training is an integral part of maintaining the highest standard of care for the waterworks system.

- **Complete the preventative and protective strategy for watershed**
  A Risk Management Strategy for the Town's water source has been adopted. The completion of this strategy will initiate a new level of preventative measures to ensure the security and quality of the Town's water.

- **Complete Standard Operating Procedures Manual**
  Consistent procedures and guidelines will provide continuity and preparedness in times of transition. The manual will work as a living document that can be updated as new knowledge is gained. It will act as a corporate memory to ensure high quality standards and effective succession planning.

- **Moving forward with Long Term Planning Initiatives**
  Continuation of steps outlined in long term planning, such as Unidirectional Flushing program, Watershed Protection Plan, Preliminary pipeline design from Holland Lake to Stocking Lake, and the feasibility of bringing water to Ladysmith from the Cassidy Aquifer.

- **Internal Communications Strategy**
  This strategy will be created to establish enhanced internal communications to the various departments for awareness, understanding and support of inter-departmental objectives for the water system. This strategy will discuss email, cell-phone, monthly reporting, and other communications and how they will be utilized in this internal communications strategy.

Chapter Five

GOALS FOR THE FUTURE

- **Continuation of the water main replacement program**
  Over the past few years, Council has approved the replacement and upgrade of the water distribution system. Crews have replaced between 400 and 600 metres of water main annually. In 2011, the section proposed to be replaced and/or upgraded is the watermain on Symonds Street between Second Avenue and Fourth Avenue.

- **Upgrading and replacing the supply lines**
  Work is underway to construct a new supply line from Holland Lake to Stocking Lake. This will allow the Town to obtain an adequate supply of water from the Holland Lake system during times of high turbidity and/or color within the watershed.

- **Centralizing the treatment system**
  The replacement of the existing raw water supply line from the south end chlorinator to the Arbutus Reservoir and the installation of a new potable water line in the same trench, will allow for a centralized treatment system that will reduce future operations costs.

- **Planning for generations**
  Tying into the Cassidy Aquifer will ensure a safe and adequate water supply for future generations.

Chapter Five

GLOSSARY

E. Coli (Escherichia coli) - are bacteria present in the intestine and feces of warm-blooded animals. E. coli are a member species of the fecal coliform group of indicator bacteria. Their concentrations are expressed as number of colonies per 100 ml of sample.

Fecal Coliform- A sub group of total coliform bacteria that are distinguished in laboratory tests at elevated temperatures. It is used as an indicator organism for potential waterborne disease risk.

Filtration - The process of passing water through a porous substance to remove solids in suspension, such as organics. Effectiveness is measured as a reduction of turbidity.

Free Chlorine - the quantity of chlorine remaining which has not been consumed in reactions with microorganisms or organic matter. Also referred to as residual chlorine.

NTU (Nephelometric Turbidity Units) - The standard unit of measurement for turbidity in water. It detects the amount of light that is scattered by finely suspended particles in water.

ved from rocks and minerals. Organic matter is characterized by its carbon-hydrogen structure.

pH - the expression of the acidity of a solution by the negative logarithm of the hydrogen ion concentration; pH of 1 is very acidic; pH of 14, very basic; pH 7 is neutral. The neutral point of 7 indicates the presence of equal concentrations of free hydrogen and free hydroxide ions.

Potable Water - Water which is considered safe and fit for human consumption, culinary and domestic purposes and meets the requirements of the health authority having jurisdiction, which is the Vancouver Island Health Authority in this region.

Pressure Zone - A zone where the pressure is constant. Pumps and valves in the distribution system, keep the pressure constant in each pressure zone.

PRV - Pressure reducing valve.

PVC - A type of plastic used to make pipes.

Raw Water - Untreated water from wells, surface sources or any water before it reaches a water treatment device or process.

Reservoir - A receptacle used for storing water within the distribution system.

Residual Chlorine - the quantity of chlorine remaining which has not been consumed in reactions with micro-organisms or organic matter. Also referred to as free chlorine.

Total Chlorine -The total amount of chlorine in a solution, which includes the chlorine demand as well as the residual chlorine.

Total Coliform - An indicator group of organisms mostly of intestinal origin used to appraise the microbiological risk to drinking water.

Turbidity - The measurement of how cloudy or murky water is. Its measurement relates to the optical property of water that causes light to be scattered and absorbed in a cloudy sample rather than transmitted in a straight line.

APPENDIX 'A'

Water Sample Range Report

Vancouver Island Health Authority

Central Island

Facility Name: TOWN OF LADYSMITH WATER WORKS
Facility Type: DWT
Date Range: Jan 1 2010 to Dec 31 2010
Date Created: Feb 23 2011

Sampling Site	Date Collected	Total Coliform	E. Coli	Fecal Coliform
606 OAKWOOD, Dist. site, Monthly				
	01/06/2010	L1	L1	
	02/02/2010	L1	L1	
	03/03/2010	L1	L1	
	04/07/2010	L1	L1	
	05/04/2010	L1	L1	
	06/01/2010	L1	L1	
	07/14/2010	L1	L1	
	08/11/2010	L1	L1	
	09/14/2010	L1	L1	
	10/05/2010	L1	L1	
	11/03/2010	L1	L1	
	12/01/2010	L1	L1	
	Total Positive:	0	0	0
Oyster Cove Road, Ladysmith, B.C., Wastewater Treatment Plant, Dist. site, Monthly				
	01/12/2010	L1	L1	
	02/16/2010	L1	L1	
	03/16/2010	L1	L1	
	04/14/2010	L1	L1	
	05/04/2010	L1	L1	
	06/01/2010	L1	L1	
	07/27/2010	L1	L1	
	08/24/2010	L1	L1	
	09/14/2010	L1	L1	
	10/19/2010	L1	L1	
	11/16/2010	L1	L1	
	12/14/2010	L1	L1	
	Total Positive:	0	0	0
340 6Th Avenue, Ladysmith, B.C., Public Works Yard, Dist. site, Monthly				
	01/19/2010	L1	L1	
	02/09/2010	L1	L1	
	03/16/2010	L1	L1	
	04/14/2010	L1	L1	
	05/17/2010	L1	L1	
	06/15/2010	L1	L1	
	07/21/2010	L1	L1	
	08/04/2010	L1	L1	
	09/22/2010	L1	L1	
	10/13/2010	L1	L1	

	12/0 010	L1	L1	
	Total Positive:	0	0	0
, 604 Farrell Road, Dist. site, Monthly				
	01/26/2010	L1	L1	
	02/09/2010	L1	L1	
	03/09/2010	2	L1	
	03/16/2010	L1	L1	
	04/07/2010	L1	L1	
	05/12/2010	L1	L1	
	06/01/2010	L1	L1	
	07/14/2010	L1	L1	
	08/11/2010	L1	L1	
	09/22/2010	L1	L1	
	10/05/2010	L1	L1	
	11/08/2010	L1	L1	
	12/07/2010	L1	L1	
	Total Positive:	1	0	0
, 405 Blair Place, Dist. site, Monthly				
	01/06/2010	L1	L1	
	02/09/2010	L1	L1	
	03/09/2010	L1	L1	
	04/07/2010	L1	L1	
	05/12/2010	L1	L1	
	06/29/2010	L1	L1	
	07/27/2010	L1	L1	
	08/31/2010	L1	L1	
	09/14/2010	L1	L1	
	10/19/2010	L1	L1	
	11/16/2010	L1	L1	
	12/01/2010	L1	L1	
	Total Positive:	0	0	0
, 1280 Rocky Creek Road, Dist. site, Monthly				
	01/12/2010	L1	L1	
	02/23/2010	L1	L1	
	03/24/2010	L1	L1	
	04/14/2010	L1	L1	
	05/04/2010	L1	L1	
	06/23/2010	L1	L1	
	07/21/2010	L1	L1	
	08/17/2010	L1	L1	
	09/28/2010	L1	L1	
	10/26/2010	L1	L1	
	11/23/2010	L1	L1	
	12/14/2010	L1	L1	
	Total Positive:	0	0	0
Ladysmith, 558 Hooper Place, Dist. site, Monthly				
	01/12/2010	L1	L1	
	02/23/2010	L1	L1	
	03/24/2010	L1	L1	
	04/21/2010	L1	L1	

06/17/2010	L1	L1	
06/2 010	L1	L1	
07/06/2010	L1	L1	
08/17/2010	L1	L1	
09/28/2010	L1	L1	
10/26/2010	L1	L1	
11/23/2010	L1	L1	
12/14/2010	7	L1	
12/20/2010	L1	L1	
Total Positive:	1	0	0

, Kinsmen Park 1000 Colonia Drive, Dist. site, Monthly

01/06/2010	L1	L1	
02/16/2010	L1	L1	
03/03/2010	L1	L1	
04/21/2010	L1	L1	
05/26/2010	L1	L1	
06/15/2010	L1	L1	
07/06/2010	L1	L1	
08/24/2010	L1	L1	
09/08/2010	L1	L1	
10/05/2010	L1	L1	
11/16/2010	L1	L1	
12/07/2010	L1	L1	
Total Positive:	0	0	0

, City Hall 410 Esplanade , Dist site, Monthly

01/26/2010	L1	L1	
02/16/2010	L1	L1	
03/30/2010	L1	L1	
04/27/2010	L1	L1	
05/26/2010	L1	L1	
06/09/2010	L1	L1	
07/06/2010	L1	L1	
08/04/2010	L1	L1	
09/22/2010	L1	L1	
10/13/2010	L1	L1	
11/08/2010	L1	L1	
12/01/2010	L1	L1	
Total Positive:	0	0	0

, 432 DAVIS ROAD, Dist site, Monthly

01/19/2010	L1	L1	
02/02/2010	L1	L1	
03/30/2010	L1	L1	
04/27/2010	L1	L1	
05/26/2010	L1	L1	
06/09/2010	L1	L1	
07/27/2010	L1	L1	
08/11/2010	L1	L1	
09/08/2010	L1	L1	
10/13/2010	L1	L1	
11/03/2010	L1	L1	
12/07/2010	L1	L1	
Total Positive:	0	0	0

LADYSMITH, TOWN OF WATER WORKS

Dist. site Annually

	11/09/2020 00	L 1	L 1	
Total Positive:	0	0	0	0

Result Values: E- estimated L- less than G- greater than

Samples that contain total coliform:	2	1.63% of total
Samples that contain e. coli:	0	0.00% of total
Samples that contain fecal coliform	0	0.00% of total
Number of positive samples in last 30 days	1/77	
Total number of samples:	123	

Comments:

two positive total coliform results out of 123 samples. Bacteriologically satisfactory results - meets Drinking Water Protection Act & Regulations Schedule A.

Environmental Health Officer
Feb 23 2011

FOR FURTHER INFORMATION PLEASE CALL: Lucko, Jill (250) 755-6215 Nanaimo

Operator

Town Of Ladysmith
Box 220 STN MAIN
Ladysmith, BC
V9G 1A2

(250) 245-6400

Water Sample Range Report
Vancouver Island Health Authority
Central Island

Facility Name: TOWN OF LADYSMITH WATER WORKS
Facility Type: DWT
Date Range: Jan 1 2010 to Dec 31 2010
Date Created: Feb 23 2011

Sampling Site	Date Collected	Total Coliform	E. Coli	Fecal Coliform
Holland Creek, Ladysmith, B.C., Chicken Ladder Dam RAW WATER, Source site, Quarterly				
	03/30/2010	51.2	1.0	
	06/15/2010	88.2	5.2	
	09/27/2010	816.4	6.3	
	11/30/2010	1299.7	52.9	
	Total Positive:	4	4	0

Result Values: E - estimated L - less than G - greater than

Samples that contain total coliform:	44	100.00% of total
Samples that contain e. coli:	44	100.00% of total
Samples that contain fecal coliform:	0	0.00% of total
Number of positive samples in last 30 days:	0/0	
Total number of samples:	44	

Comments:

Raw water results

Environmental Health Officer
Feb 23 2011

FOR FURTHER INFORMATION PLEASE CALL: Lucko, Jill (250) 755-6215 Nanaimo

Operator

Town Of Ladysmith
Box 220 STN MAIN
Ladysmith, BC
V9G 1A2

(250) 245-6400

Water Sample Range Report

Vancouver Island Health Authority
Central Island

Facility Name: TOWN OF LADYSMITH WATER WORKS
Facility Type: DWT
Date Range: Jan 1 2010 to Dec 31 2010
Date Created: Feb 23 2011

Sampling Site	Date Collected	Total Coliform	E. Coli	Fecal Coliform
Lake, Stocking Lake RAW WATER, Source site, Quarterly				
	03/30/2010	18.9	1.0	
	06/15/2010	21.1	L1	
	09/27/2010	198.9	1.0	
	11/30/2010	49.6	L1	
	Total Positive:	4	2	0

Result Values: E - estimated L - less than G - greater than

Samples that contain total coliform:	4	100.00% of total
Samples that contain e. coli:	2	50.00% of total
Samples that contain fecal coliform:	0	0.00% of total
Number of positive samples in last 30 days:	0/0	
Total number of samples:	4	

Comments:

Raw water results.

Environmental Health Officer
Feb 23 2011

FOR FURTHER INFORMATION PLEASE CALL: Lucko, Jill (250) 755-6215 Nanaimo

Operator

Town Of Ladysmith
Box 220 STN MAIN
Ladysmith, BC
V9G 1A2

(250) 245-6400

APPENDIX 'B'

Ladysmith, Town of
Box 220
Ladysmith, BC
V9G 1A2

TEL: (250) 245-3995
FAX: (250) 245-3711
cbaker@ladysmith.ca.

23Mar10 2:49p W88416
FWS
water
2

Arrival temp.: 6.0C
Sampler: Perry Madsen

Sample: 1)1280 Rocky Creek Rd 23Mar10 08:10a 2)558 Hooper Pl 23Mar10 08:30a

TRIHALOMETHANES

COMPOUNDS	1 SAMPLE	2 SAMPLE	UNITS
1) Bromodichloromethane	1.45	<0.100	ug/L
2) Bromoform	<0.100	<0.100	ug/L
3) Chloroform	41.5	47.0	ug/L
4) Dibromochloromethane	<0.100	<0.100	ug/L
SURROGATE RECOVERY			
1) Bromofluorobenzene	91.5	107	%
2) Toluene-d8	88.2	99.8	%

Comment: maximum allowable concentration of Total Trihalomethanes permitted is 100 ug/L (0.1 mg/L); Guidelines for Canadian Drinking Water Quality, 6th edition, 1996.

ND = not detected
ug/L = micrograms per liter
< = less than

R. Bilodeau
Analytical Chemist

H. Hartmann
Sr.Analytical Chemist

M.B. LABS LTD
T: 656-1334 F: 656-0443 E: mblabs@pacificcoast.net W: www.mblabs.com

Ladysmith, Town of
Box 220
Ladysmith, BC
V9G 1A2

22Jun10 2:23p W89920
FWS
water
2

TEL: (250) 245-3995
FAX: (250) 245-3711
cbaker@ladysmith.ca.

Arrival temp.: 6.0C
Sampler: P. Madsen

Samples: 1) 1280 Rocky Creek Rd 22Jun10 07:50a 2) 558 Hooper Pl. 22Jun10 08:23a

TRIHALOMETHANES

COMPOUNDS	1 SAMPLE	2 SAMPLE	UNITS
1) Bromodichloromethane	3.16	3.28	ug/L
2) Bromoform	<0.100	<0.100	ug/L
3) Chloroform	47.8	53.2	ug/L
4) Dibromochloromethane	<0.100	<0.100	ug/L
SURROGATE RECOVERY			
1) Bromofluorobenzene	106	112	%
2) Toluene-d8	97.2	90.1	%

Comment: maximum allowable concentration of Total Trihalomethanes permitted is 100 ug/L (0.1 mg/L); Guidelines for Canadian Drinking Water Quality, 6th edition, 1996.

ND = not detected
ug/L = micrograms per liter
< = less than

R. Bilodeau
Analytical Chemist

H. Hartmann
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Ladysmith, Town of
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V9G 1A2

21Jul10 1:58p W90397
FWS
water
2

TEL: (250) 245-3995
FAX: (250) 245-3711
cbaker@ladysmith.ca.

Arrival temp.: 5.0C
Sampler: A. Schade

SAMPLE	DATE	TIME	MCAA (ug/L)	DCAA (ug/L)	MBAA (ug/L)	DBAA (ug/L)	BCAA (ug/L)
1280 Rocky Creek Rd.	21Jul10	08:30a	ND	17.8	ND	37.4	ND
558 Hooper Pl.	21Jul10	08:20a	ND	16.2	ND	45.2	ND
Lab Blank			ND	ND	ND	ND	ND
So			0.030	0.030	0.030	0.030	0.030
REF. VALUE			10.0	10.0	10.0	10.0	10.0
STD ñ 2SD			9.87 ñ 1.05	9.90 ñ 1.04	9.91 ñ 1.01	9.89 ñ 0.991	9.90 ñ 1.01

SAMPLE	DATE	TIME	TCAA (ug/L)
1280 Rocky Creek Rd.	21Jul10	08:30a	36.4
558 Hooper Pl.	21Jul10	08:20a	34.8
Lab Blank			ND
So			0.050
REF. VALUE			10.0
STD ñ 2SD			9.87 ñ 1.01

MCAA = Monochloro Acetic Acid
MBAA = Mono2bromo Acetic Acid
DCAA = Dichloro Acetic Acid
BCAA = Bromochloro Acetic Acid
DBAA = Dibromo Acetic Acid
TCAA = Trichloro Acetic Acid

Comment: maximum allowable concentration of Total Haloacetic acids (HAA5) permitted is 60.0 ug/L (0.06 mg/L); EPA 816-F-01-010 May 2002.

ND = not detected
ug/L = micrograms per liter
< = less than

R. Bilodeau
Analytical Chemist

H. Hartmann
Sr.Analytical Chemist

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TEL: (250) 245-3995
FAX: (250) 245-3711
cbaker@ladysmith.ca.

29Sep10 2:17p W91657 pg2
FWS
water
2

Arrival temp.: 6.0C
Sampler: P. Madsen

Samples: -1) 1280 Rocky Creek Rd 29Sep10 07:52a 2) 558 Hooper Place 29Sep10 08:30a

TRIHALOMETHANES

COMPOUNDS	1 SAMPLE	2 SAMPLE	UNITS
1) Bromodichloromethane	<0.100	<0.100	ug/L
2) Bromoform	<0.100	<0.100	ug/L
3) Chloroform	69.3	72.0	ug/L
4) Dibromochloromethane	<0.100	<0.100	ug/L
SURROGATE RECOVERY			
1) Bromofluorobenzene	95.1	96.0	%
2) Toluene-d8	101	104	%

Comment: maximum allowable concentration of Total Trihalomethanes permitted is 100 ug/L (0.1 mg/L); Guidelines for Canadian Drinking Water Quality, 6th edition, 1996.

ND = not detected
ug/L = micrograms per liter
< = less than

R. Bilodeau
Analytical Chemist

H. Hartmann
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Ladysmith, Town of
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V9G 1A2

29Sep10 2:17p W91657
FWS
water
2

TEL: (250) 245-3995
FAX: (250) 245-3711
cbaker@ladysmith.ca.

Arrival temp.: 6.0C
Sampler: P. Madsen

SAMPLE	DATE	TIME	MCAA (ug/L)	DCAA (ug/L)	MBAA (ug/L)	DBAA (ug/L)	BCAA (ug/L)
1280 Rocky Creek Rd	29Sep10	07:52a	ND	26.7	ND	ND	ND
558 Hooper Place	29Sep10	08:30a	ND	15.3	ND	ND	ND
Lab Blank			ND	ND	ND	ND	ND
So			0.030	0.030	0.030	0.030	0.030
REF. VALUE			10.0	10.0	10.0	10.0	10.0
STD ñ 2SD			9.87 ñ 1.04	9.90 ñ 1.03	9.91 ñ 0.999	9.89 ñ 0.980	9.90 ñ 1.00

SAMPLE	DATE	TIME	TCAA (ug/L)
1280 Rocky Creek Rd	29Sep10	07:52a	43.8
558 Hooper Place	29Sep10	08:30a	50.4
Lab Blank			ND
So			0.050
REF. VALUE			10.0
STD ñ 2SD			9.87 ñ 0.999

SD = standard deviation
STD = secondary standard calibrated to primary standard reference material
So = standard deviation at zero analyte concentration; method detection limit is generally considered to be 3x So value
ND = none detected n/a = not applicable

MCAA = Monochloro Acetic Acid
MBAA = Mono2bromo Acetic Acid
DCAA = Dichloro Acetic Acid
BCAA = Bromochloro Acetic Acid
DBAA = Dibromo Acetic Acid
TCAA = Trichloro Acetic Acid

Comment: maximum allowable concentration of Total Haloacetic acids (HAA5) permitted is 60.0 ug/L (0.06 mg/L); EPA 816-F-01-010 May 2002.

ND = not detected
ug/L = micrograms per liter
< = less than

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Ladysmith, Town of
Box 220
Ladysmith, BC
V9G 1A2

15Dec10 2:05p W92862
FWS
water
2

TEL: (250) 245-3995
FAX: (250) 245-3711
cbaker@ladysmith.ca.

Arrival temp.: 6.0C
Sampler: Arnold

Samples: Quarterly

SAMPLE	DATE	TIME	MCAA (ug/L)	DCAA (ug/L)	MBAA (ug/L)	DBAA (ug/L)	BCAA (ug/L)
558 Hooper Pl.	15Dec10	08:00a	ND	ND	ND	3.09	ND
1280 Rocky Creek Rd	15Dec10	08:30a	ND	ND	ND	3.61	ND
1280 Rocky Creek Rd	DUP	08:30a	ND	ND	ND	4.19	ND
Lab Blank			ND	ND	ND	ND	ND
Se			0.030	0.030	0.030	0.030	0.030
REF. VALUE			10.0	10.0	10.0	10.0	10.0
STD ñ 2SD			9.87 ñ 1.04	9.90 ñ 1.02	9.91 ñ 0.992	9.89 ñ 0.973	9.91 ñ 0.995

SAMPLE	DATE	TIME	TCAA (ug/L)
558 Hooper Pl.	15Dec10	08:00a	45.3
1280 Rocky Creek Rd	15Dec10	08:30a	44.6
1280 Rocky Creek Rd	DUP	08:30a	46.6
Lab Blank			ND
Se			0.050
REF. VALUE			10.0
STD ñ 2SD			9.88 ñ 0.992

MCAA = Monochloro Acetic Acid MBAA = Mono2bromo Acetic Acid
DCAA = Dichloro Acetic Acid BCAA = Bromochloro Acetic Acid
DBAA = Dibromo Acetic Acid TCAA = Trichloro Acetic Acid

Comment: maximum allowable concentration of Total Haloacetic acids (HAA5) permitted is 60.0 ug/L (0.06 mg/L); EPA 816-F-01-010 May 2002.

ND = not detected
ug/L = micrograms per liter
< = less than

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V9G 1A2

TEL: (250) 245-3995
FAX: (250) 245-3711
cbaker@ladysmith.ca.

15Dec10 2:05p W92862 pg2
FWS
water
2

Arrival temp.: 6.0C
Sampler: Arnold

Samples: Quarterly 1) 558 Hooper Pl 15Dec10 08:00a 2) 1280 Rocky Creek Rd 15Dec10 08:30a

TRIHALOMETHANES

COMPOUNDS	1 SAMPLE	2 SAMPLE	UNITS
1) Bromodichloromethane	<0.100	<0.100	ug/L
2) Bromoform	<0.100	<0.100	ug/L
3) Chloroform	49.0	48.1	ug/L
4) Dibromochloromethane	<0.100	<0.100	ug/L
SURROGATE RECOVERY			
1) Bromofluorobenzene	99.5	108	%
2) Toluene-d8	103	112	%

Comment: maximum allowable concentration of Total Trihalomethanes permitted is 100 ug/L (0.1 mg/L); Guidelines for Canadian Drinking Water Quality, 6th edition, 1996.

ND = not detected
ug/L = micrograms per liter
< = less than

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Ladysmith, Town of
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Ladysmith, BC
V9G 1A2

29Jul10 3:12p W90573
FWS
water
3

TEL: (250) 245-3995
cbaker@ladysmith.ca.

Arrival temp.: 4.0C

Samples: 1) Holland Lake 29Jul10 08:30a 2) Stocking Lake 29Jul10 08:25a
3) Chicken Ladder 29Jul10 09:05a

ELEMENTS		1 SAMPLE	2 SAMPLE	3 SAMPLE	UNITS	Max.for Aquatic Life Freshwater	Marine	Maximum Permissible In Effluent **
1) Aluminium	Al	<0.065	<0.065	<0.065	mg/L	n/a	1.50	4.00
2) Antimony	Sb	<0.500	<0.500	<0.500	ug/L	n/a	200	5.00
3) Arsenic	As	<0.500	<0.500	<0.500	ug/L	5.00	12.5	250
4) Barium	Ba	<0.009	<0.009	<0.009	mg/L	n/a	1.00	1.00
5) Beryllium	Be	<0.003	<0.003	<0.003	mg/L	0.010	1.50	no limit listed
6) Boron	B	<0.050	<0.050	<0.050	mg/L	n/a	5.00	5.00
7) Cadmium	Cd	0.415	<0.100	0.201	ug/L	1.05-2.11	0.12	10.0
8) Calcium	Ca	1.11	2.95	1.59	mg/L	n/a	n/a	no limit listed
9) Chromium	Cr	<0.010	<0.010	<0.010	mg/L	0.100	0.050	0.030
10) Cobalt	Co	<0.020	<0.020	<0.020	mg/L	1.32	n/a	0.500
11) Copper	Cu	<0.008	<0.008	<0.008	mg/L	0.030-0.127	0.050	0.500
12) Gold	Au	<0.040	<0.040	<0.040	mg/L	n/a	n/a	no limit listed
13) Iron	Fe	0.074	0.034	0.036	mg/L	1.00	0.300	1.00
14) Lanthanum	La	<0.020	<0.020	<0.020	mg/L	n/a	n/a	no limit listed
15) Lead	Pb	<0.500	<0.500	<0.500	ug/L	30.0	50.0	100
16) Magnesium	Mg	0.183	0.382	0.285	mg/L	n/a	n/a	no limit listed
17) Manganese	Mn	0.010	<0.004	<0.004	mg/L	n/a	0.100	0.050
18) Mercury	Hg	<0.100	<0.100	<0.100	ug/L	2.00	1.00	5.00
19) Molybdenum	Mo	<0.020	<0.020	<0.020	mg/L	n/a	n/a	0.500
20) Nickel	Ni	<0.050	<0.050	<0.050	mg/L	n/a	0.100	0.500
21) Phosphorus	P	<0.065	<0.065	<0.065	mg/L	n/a	0.050	1.50
22) Potassium	K	<0.030	0.036	<0.030	mg/L	n/a	n/a	no limit listed
23) Scandium	Sc	<0.050	<0.050	<0.050	mg/L	n/a	n/a	no limit listed
24) Selenium	Se	<0.500	<0.500	<0.500	ug/L	10.0	10.0	100
25) Silicon	Si	0.099	0.237	0.186	mg/L	n/a	n/a	no limit listed
26) Silver	Ag	<0.010	<0.010	<0.010	mg/L	0.010	0.005	1.00
27) Sodium	Na	<0.050	0.616	0.377	mg/L	n/a	n/a	no limit listed
28) Strontium	Sr	0.005	0.009	0.008	mg/L	75.0	n/a	no limit listed
29) Tin	Sn	<0.020	<0.020	<0.020	mg/L	n/a	n/a	no limit listed
30) Titanium	Ti	<0.010	<0.010	<0.010	mg/L	n/a	n/a	no limit listed
31) Tungsten	W	<0.050	<0.050	<0.050	mg/L	n/a	n/a	no limit listed
32) Vanadium	V	<0.010	<0.010	<0.010	mg/L	n/a	10.0	no limit listed
33) Zinc	Zn	<0.001	<0.001	<0.001	mg/L	0.490-1.35	0.100	5.00
Hardness (mg/L CaCO3)		3.51	8.94	5.14	mg/L	0-75 mg/L = soft		
pH		6.72	7.08	6.93	units	6.5-9.0	6.5-9.0	5.5-11.0

As per Canadian or B.C. limits Ministry of Environment - Water Quality Criteria, Report No. 80-9, 1980. Task Force of the Canadian Council of Resource & Envir. Ministers
As per Canadian Environmental Quality Guidelines Summary Table. Updated Dec 2003 Chapter 4, Water: Aquatic Life. EPA-822-R-02-07 & US EPA 822/R-85-100; WHO Eniv.134 1992

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APPENDIX 'C'

Town of Ladysmith ANNUAL REPORT

(Drinking Water System Name)

Reporting Period:	January 1, 2010 – December 31, 2010
Operating Permit Number:	759
Drinking Water System Owner:	Town of Ladysmith
Drinking Water System Contact: Name: Curtis Baker Phone No: (250) 245-3995 Email: cbaker@ladysmith.ca	

1 **Microbiological testing completed during this reporting period:**

a. bacteriological results attached to this report.

b. adverse bacteriological results: None detected

x Listed in table below:

Adverse Results:

Date	Total coliform	E. Coli	Reason	Corrective Action
12/14/10	7		Sampling error	Re-sampled
03/09/10	2		Sampling error	Re-sampled

2 **Chemical results for this reporting period:**

a. most recent chemical analysis attached to this report.

b. chemical parameters listed in *The Guidelines for Canadian Drinking Water Quality ("the Guidelines")* are:

X all within Guidelines

☐ above the Guidelines and are listed below:

Parameters above the Guidelines:

Parameter	Result	Max. Acceptable Concentration	Aesthetic Objective	Treatment/Corrective Action

Town of Ladysmith ANNUAL REPORT
(Drinking Water System Name)

3 Summarize additional testing and sampling carried out in accordance with the requirement of a Water Source approval, Written Order or as per the conditions of your *Operating Permit*.

X no additional testing
☐ additional testing listed below:

Additional testing:

Description of parameter & reason for sampling	Health parameter or non-health related parameter	Corrective action necessary (Y/N?)	Corrective action taken

4 Water Quality Complaints:

During the course of the year, the water system:
X did not receive water quality complaints (ie taste, odour, colour, etc)
☐ received water quality complaints and are listed below:

Water Quality Complaints:

Date	Water quality complaint	Corrective action taken

5 Adverse results: Total number of adverse results during this reporting period for insufficient water supply, malfunction of disinfection equipment or elevated turbidity:

No adverse results
X Adverse results listed below:

Adverse Results:

Incident date	Corrective action	Corrected by
06/24/10	Malfunction of valve (fixed)	Utilities Operators

Town of Ladysmith ANNUAL REPORT
(Drinking Water System Name)

6 Description of the system:

Sources of raw water:

- ☐ Groundwater
- X Surface water
- ☐ Other (specify): _____

Does the drinking water system have disinfection? X Yes ☐ No

Disinfection methods (check boxes that apply):

- X Chlorination
- ☐ Ultraviolet light
- ☐ Ozonation
- ☐ Other (specify): _____

Does the drinking water system have treatment? ☐ Yes X No

Treatment type (check boxes that apply):

- ☐ Particulate cartridge filters
- ☐ Membrane filtration
- ☐ Carbon filter
- ☐ Sand filtration
- ☐ Reverse osmosis
- ☐ Other (specify): _____

7 Major expenses incurred during the period covered by the report:

To purchase or install required equipment: $69,825

To repair equipment: _____

To replace equipment: _____

To complete annual maintenance of system: *(system flushing, replacement of carbon filters, etc)* $453,373

To complete specialist report (specify): _____

8 Further communication with users:

a. Indicate how you notified system users that your annual report is available, and is free of charge:

- ☐ hand delivered
- X public access/ notice via web
- ☐ public access/notice via government office
- ☐ public access/notice via newspaper
- ☐ public access/notice via bill stuffer
- ☐ public access/ notice via other method (specify): _____

b. Improvements or remedial actions required by the Drinking Water Officer:
- X no action required
- ☐ Drinking Water Officer inspection report attached to report
- ☐ actions required by Drinking Water Officer listed below:

Improvements/Remedial Actions:

Required action	Completion date

c. Future water system improvements:
- ☐ no improvements planned
- X improvements listed below:

Future Improvements:

Future plans	Planned completion date
Centralized Treatment Facility	March 31, 2012
Interconnecting Pipeline	December 31, 2011

d. Emergency Response Plan can be accessed by:
- X posting on web
- ☐ posting at nearest government office
- ☐ contacting water system owner
- ☐ Other (specify): _____

Town of Ladysmith

STAFF REPORT

To: Ruth Malli, City Manager
From: Joe Friesenhan, Director of Public Works
Date: June 29, 2011
File No:

Re: LOW CONSUMPTION TOILETS

RECOMMENDATION(S):
That the Committee recommend that Council direct staff to prepare an amendment to Section 6.12, Water Conservation, of the Town of Ladysmith Building and Plumbing Bylaw 1994, No 1119 to reduce the average water consumption of a water closet or direct flush urinals from 6 litres per flush cycle to 4.8 litres per flush cycle, and from 3.8 litres per flush cycle to 1.9 litres per flush cycle respectively.

PURPOSE:
To update the Town of Ladysmith Building and Plumbing Bylaw 1994, to meet the new plumbing and fixture requirements in the BC Building Code.

INTRODUCTION/BACKGROUND:
In 2004, the Town amended the Building and Plumbing Bylaw 1994, No. 1119 by adding the following section:

6.12 Water Conservation

6.12.1 Every water closet, whether tank type or dual flush, must be a low consumption type that

(a) conforms with the latest Canadian Standard Association (CSA) standards, and
(b) has an average water consumption not exceeding 6 litres per flush cycle.

6.12.2 Direct flush urinals must be a low consumption type that

(a) conforms with the latest Canadian Standards Association (CSA) standards and
(b) has an average water consumption not exceeding 3.8 litres per flush cycle

The Province has recently announced that it is taking another step toward increased water efficiency through new high efficiency toilet and urinal requirements in the BC Building Code. Effective October 3, 2011, high efficiency toilets or dual flush toilets will be required in new residential buildings or when renovations involving plumbing fixtures occur. Whenever urinals are installed, high efficiency urinals will be required.

In order to meet the new requirements of the Building Code, the following addition to the Town of Ladysmith Building and Plumbing Bylaw 1994, No 1119 Amendment Bylaw (No. 11) 2004, No. 1551 is required:

6.12.1 (b) has an average water consumption not exceeding 4.8 litres per flush, and
6.12.2 (b) has an average water consumption not exceeding 1.9 litres per flush

SCOPE OF WORK:
N/A

ALTERNATIVES:
N/A

FINANCIAL IMPLICATIONS:
N/A

LEGAL IMPLICATIONS:
This bylaw amendment is required in order to maintain compliance with the BC Building Code.

CITIZEN/PUBLIC RELATIONS IMPLICATIONS:
Continued reduction in water consumption is consistent with the Town's sustainability initiatives and should be positively received by the general public. Builders/renovators could express concern about potential higher costs of the higher-efficiency fixtures.

INTERDEPARTMENTAL INVOLVEMENT/IMPLICATIONS:
N/A

RESOURCE IMPLICATIONS:
N/A

ALIGNMENT WITH SUSTAINABILITY VISIONING REPORT:
Aligns with Sustainability Pillars #3, Green Buildings, and #5, Innovative Infrastructure.

ALIGNMENT WITH STRATEGIC PRIORITIES:
The proposed bylaw amendment aligns with Strategic Direction D – Enhanced Standard of Infrastructure – Ensure a safe and sustainable water supply.

SUMMARY:
In order to meet the new requirements of the BC Building Code, which will come into effect on October 3, 2011, the following additions to the Building and Plumbing Bylaw 1994, 1119 are required:

6.12.1 (b) has an average water consumption not exceeding 4.8 litres and
6.12.2 (b) has an average water consumption net exceeding 1.9 litres

I concur with the recommendation.

Ruth Malli, City Manager

ATTACHMENTS:
"None".